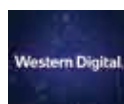




Samsung X5 Portable SSD

Samsung's first NVMe-based portable SSD offering up to 40 Gbps bandwidth has finally launched in India. <http://dgit.in/feb21-73>



WD Purple Expands Storage

Western Digital expands purple surveillance storage options with 18TB HDD and 1TB microSDXC models. <http://dgit.in/feb21-74>

WD Black SN850 NVMe SSD 1 TB

Blazing fast



Price
₹20,649

The WD SN850 is the latest addition to the WD Black family of storage media using the M.2 form factor. However, the more important distinction is that the SN850 is the company's first PCIe Gen 4.0 drive. So there's a lot of stuff that's new with this SSD. WD has been using their own custom controller which they've kept under wraps for a while now. And the NANDs used are also a little denser than what the previous gen SN750 had under the hood.

The SSD now boasts of sequential read speeds of about 7000 MBps, enough to put most of the competition to shame. However, when you push this hard, we're going to see a few chinks in the armour. And that's what we're looking for.

Build and Design

We received the non- heatsink variant of the SSD which is odd since none of the PCIe Gen 4.0 SSDs come without heatsinks. You could say that WD has made an efficient controller which can do without the heatsink. We even asked for the heatsink variant to see if that would make a difference. The SN750's heatsink had posed some fitting problems with motherboards that had their PCIe slots too close to each other with an M.2 slot between the two. For this, WD had come out with a compatibility matrix of popular motherboards. We haven't seen the heatsink this time around, so we can't say for certain how wide it is but having glanced at the press renders, there's still a metal lip on the sides which adds to the width.

The 1 TB SKU is a single-sided PCB so there are no mounting issues and if your motherboard comes with a heat-sink for the M.2 slots then that would be appropriate considering PCIe Gen 4.0 SSDs do generate quite a bit of heat.



PERFORMANCE.....87
BUILD.....70
VALUE FOR MONEY....79

The 2 TB SKU on the other hand is a double-sided PCB so you'll need a double sided heat sink solution for best results.

Performance

WD has had a custom controller for quite some time and with the last two iterations of the WD

Black NVMe SSDs, we can say that it has worked for them. Most of the PCIe Gen 4.0 SSDs that we've come across have had PHISON controllers so performance hasn't varied by that much across the board. Which makes the WD Black SN850 all the more special since Samsung is the only other major manufacturer with a custom controller. Speaking of custom controllers, the new one that the WD SN850 is using is an 8-channel PCIe 4.0 x4 controller that is connected to 256Gb dies and comes with hybrid SLC caching. We've seen how QLC NANDs use a portion of the NAND as an SLC cache to compensate for the low write performance of QLC NAND. You can think of WD's nCache 4.0 as somewhat the same. One third of the WD SN850 is treated as a hybrid cache in conjunction with the 12 GB actual SLC cache. This helps with performance and endurance of the drive.

Sequential Read Performance

The WD SN850 scores more than the

WD Black SN750 in our sustained 128K read tests thanks to this being a PCIe Gen 4.0 drive. We managed to score very close to the advertised 7000 MBps read speeds although consistency was a problem. The drive heats up a bit so when you run the benchmark for the first time, you get the advertised speeds but once the drive temperature has gone up, we see a performance hit of anywhere between 400-600 MBps. Getting the same performance again requires a good heatsink or you'll have to channel plenty of cooling over the drive to normalise temperatures.

Sequential Write Performance

Even in the sequential write performance test we find the WD SN850 to get the advertised 5300 MBps speeds and once the drive heats up, we don't see that much of a performance differential but it's very much present.

Verdict

So, the WD SN850 turned out to be a little hot-headed with some amount of thermal throttling happening. We reached out to WD and they did provide us with another sample on which we experienced the same drop in performance so if you're planning on getting the SN850, we would advise that you get a motherboard with a good heatsink since you will need it, as is the case with any of the other PCIe Gen 4.0 SSDs. Or you could simply wait and get the heatsink variant. Either way, it's a decent high-performance SSD which is a good candidate for your boot drive and for storing your massive games.

—Mithun Mohandas

SPECIFICATIONS

WD100T1X0E | FORM FACTOR: M.2 2280 | COST/GB: ₹5.20/17 | DIMENSIONS: 80 x 22 x 8.10 mm | CONTROLLER: WD Proprietary | BiCS4 96-layer TLC NAND | WARRANTY: 5 years

CONTACT

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